



AqOne

OFFLINE MARITIME SAFETY SYSTEM FOR MUNICIPAL AND SMALL-SCALE FISHERS

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ABSTRACT

AqOne is an offline-capable maritime safety prototype designed for small-scale fishers operating beyond cellular coverage. A fisher can generate a GPS-enabled SOS through the mobile application, transfer it to a boat safety pod over local Wi-Fi, and relay the message through a LoRa-based network toward an internet-connected shore gateway and responder dashboard. The system explores low-cost, store-and-forward emergency communication for coastal communities where continuous connectivity cannot be guaranteed.

PROBLEM STATEMENT

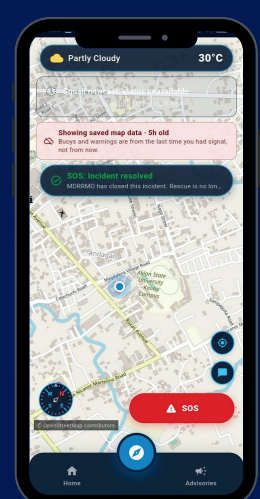
"When cellular signal ends, communication ends."



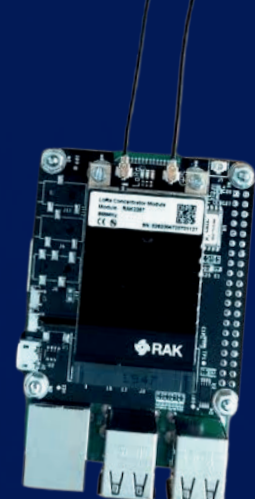
Key challenges:

- No cellular signal offshore
- No direct emergency link
- Delayed reporting of distress incidents
- No reliable location/status sharing
- Limited coordination between fishers and responders.

HOW AQONE WORKS



SEND SOS



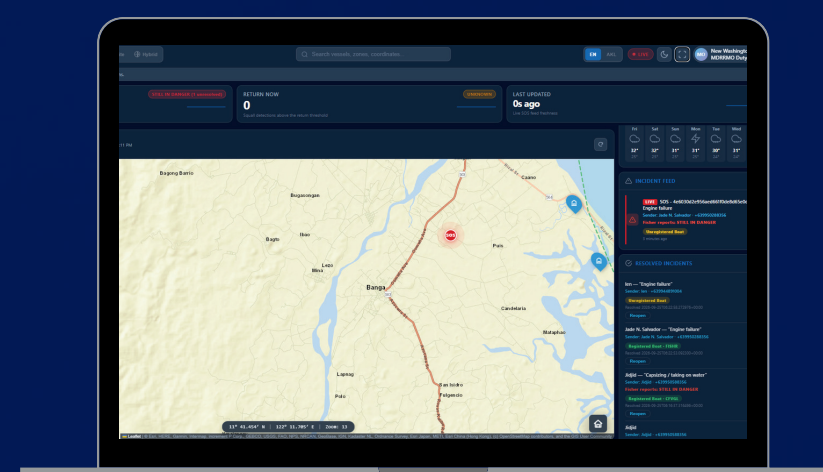
POD RECEIVES



BUOYS RELAY

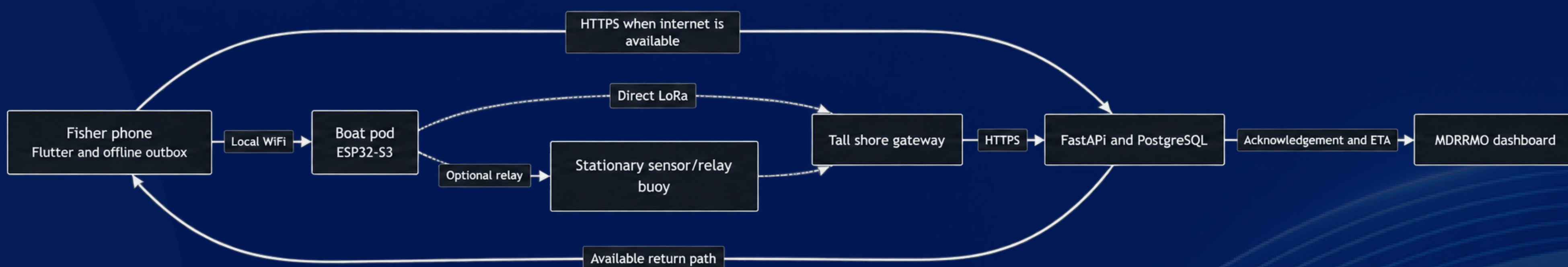


BACKEND PROCESSES

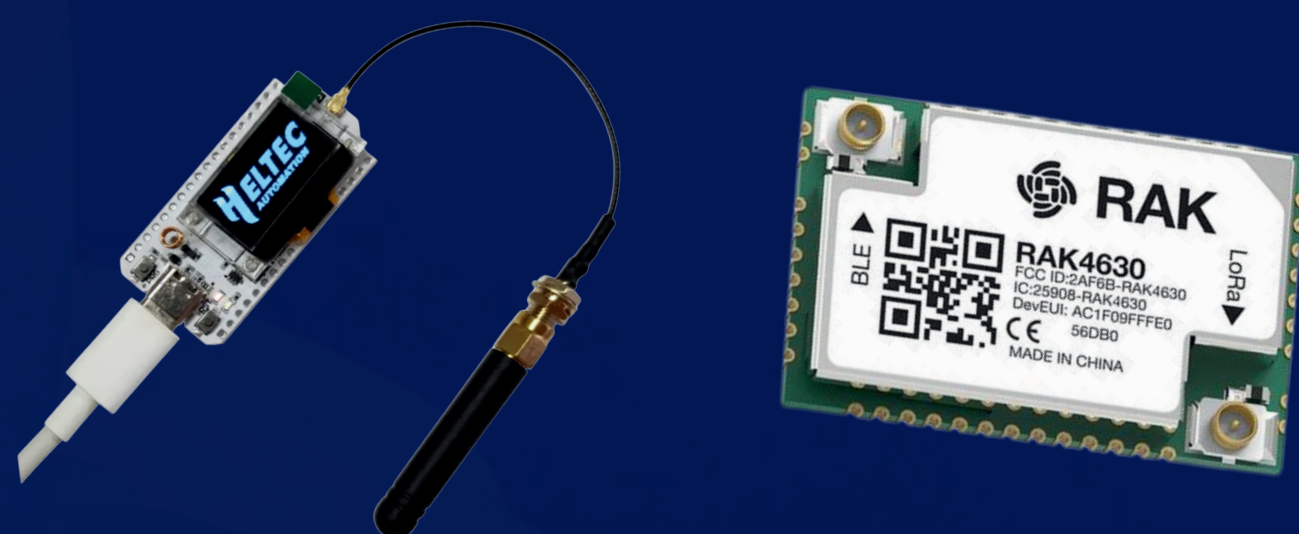


DASHBOARD ALERTS

SYSTEM ARCHITECTURE



WHY LORA



LoRa, or Long Range, is a low-power wireless technology designed to send small amounts of data over long distances. AqOne uses it to transmit SOS alerts, GPS coordinates, and device status between boats, relay nodes, and shore gateways. It helps maintain communication in areas where cellular coverage is weak or unavailable.

OBJECTIVES



- Develop an offline maritime safety system for small-scale fishers.
- Enable GPS-based SOS beyond cellular coverage.
- Relay SOS messages through LoRa to shore.
- Provide MDRMO with real-time monitoring and response tools.
- Improve emergency coordination between fishers and responders.

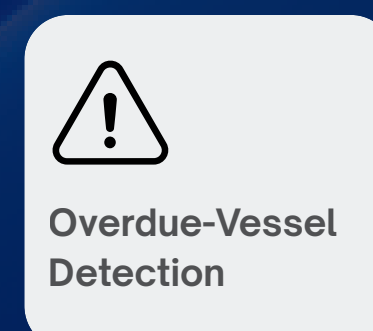
DISCUSSION & LIMITATIONS

AqOne uses store-and-forward communication, so continuous connectivity is not guaranteed. Full end-to-end field testing and outdoor range validation are still pending. Weather, overdue-trip, and drift-search features are decision-support tools and do not automatically trigger an SOS or dispatch rescue.

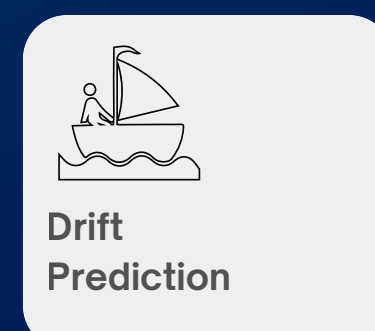
OTHER FEATURES



Weather & Hazard Intelligence



Overdue-Vessel Detection



Drift Prediction

Provides weather and hazard intelligence, overdue-vessel detection, and drift prediction to support safer trips and emergency response. These tools help monitor conditions, flag vessels that may need attention, and estimate possible drift areas for responder review.

TECHSTACK



"Gabay sa Bawat Alon, Konektado sa Bawat Layon"